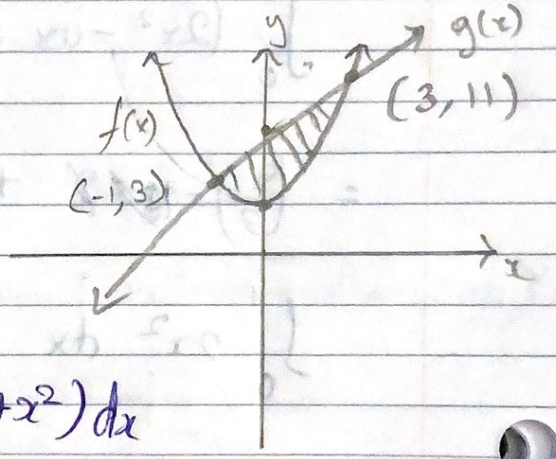
Quiz -2 :-

$f(x) = 2 + x^2$ and $g(x) = 5 + 2x$

$2 + x^2 = 5 + 2x \Rightarrow x^2 - 2x - 5 + 2 = 0$

$\Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0$

$\boxed{x = 3}$, $\boxed{x = -1}$



(A) dx :-

$$A = \int_{-1}^{3} (5 + 2x) - (2 + x^2)\, dx$$

$$A = \int_{-1}^{3} \left[(5 + 2x) - 2 - x^2 \right] dx$$

$$\boxed{A = \int_{-1}^{3} (3 + 2x - x^2)\, dx}$$

(B) dy :-

$y = 2 + x^2 \Rightarrow y - 2 = x^2 \Rightarrow \boxed{\sqrt{y-2} = x}$

$y = 5 + 2x \Rightarrow y - 5 = 2x \Rightarrow \boxed{\frac{y-5}{2} = x}$

$$\boxed{A = \int_{3}^{11} \left(\frac{y-5}{2} - \sqrt{y-2} \right) dy}$$

(c) I would choose A which is

$$A = \int_{-1}^{3} (3 + 2x - x^2)\, dx$$